

# Debt Expansion Restraint and the Cross Section of Stock Returns

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October 20, 2025

## Abstract

This paper studies the asset pricing implications of Debt Expansion Restraint (DER), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on DER achieves an annualized gross (net) Sharpe ratio of 0.48 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (19) bps/month with a t-statistic of 3.22 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets) is 22 bps/month with a t-statistic of 3.01.

# 1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, yet the cross-sectional variation in expected returns remains only partially explained by standard consumption-based models. While rational asset pricing theory predicts that firms with higher systematic risk exposure should earn higher expected returns, identifying firm characteristics that reliably capture this consumption risk exposure continues to challenge researchers. Corporate financing decisions, particularly those related to debt expansion, potentially signal managers' private information about future cash flow risks and their correlation with aggregate consumption growth. Firms that exercise restraint in debt expansion may face different exposures to consumption risk than those aggressively leveraging their balance sheets, suggesting that debt financing patterns could predict cross-sectional return differences through a risk-based channel.

Debt Expansion Restraint (DER) captures firms' conservative approach to leverage increases, which we argue reflects their exposure to consumption risk through multiple channels. First, firms exhibiting debt restraint likely face higher costs of external finance during bad economic states when marginal utility is high, making them particularly sensitive to consumption shocks ([Livdan et al., 2009](#)). These financing frictions become more binding precisely when households' marginal utility peaks, creating covariation between firm cash flows and the stochastic discount factor ([Gomes and Schmid, 2016](#)). Second, conservative debt policies may indicate managers' concerns about long-run consumption risks, as firms maintaining financial flexibility can better navigate persistent downturns that affect aggregate consumption growth ([Bansal and Yaron, 2004](#)). The state-dependent nature of financing constraints implies that debt-constrained firms experience amplified sensitivity to consumption disasters, when credit markets freeze and the representative agent's marginal utility spikes ([Barro, 2006](#)). Third, firms restraining debt expansion po-

tentially signal exposure to habit formation risks, as their conservative financing preserves options to smooth production when consumption approaches the habit level (Campbell and Cochrane, 1999). The intertemporal marginal rate of substitution prices these firms at a premium because their cash flows covary negatively with consumption innovations during periods of heightened risk aversion.

We find strong evidence that Debt Expansion Restraint predicts cross-sectional stock returns. The value-weighted long/short DER strategy achieves an annualized gross Sharpe ratio of 0.48, with monthly average returns of 26 basis points (t-statistic = 3.45). The strategy’s alpha relative to the Fama-French six-factor model equals 25 basis points per month (t-statistic = 3.22), demonstrating robustness to standard risk adjustments. Among large-cap stocks exceeding the 80th NYSE percentile, the DER strategy delivers even stronger performance with average returns of 34 basis points per month (t-statistic = 3.65), alleviating concerns about small-stock illiquidity driving results. The strategy loads significantly on the CMA factor with a coefficient of 0.24 (t-statistic = 4.66), consistent with debt-restrained firms’ investment conservatism during high marginal utility states. After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains economically meaningful at 0.37, placing DER in the top 6% of anomalies in the factor zoo. The strategy maintains statistical significance across alternative portfolio construction methods, with nineteen of twenty-five specifications producing t-statistics exceeding three for gross alphas.

Our paper contributes to the consumption-based asset pricing literature by establishing debt expansion restraint as a robust predictor of returns through its connection to consumption risk exposure. We extend (Livdan et al., 2009) by demonstrating that financing decisions predict returns not through behavioral biases but through rational pricing of state-dependent consumption risk. Unlike (Cooper et al., 2008), who focus on asset growth’s relation to expected returns through investment

frictions, we show that the debt financing component specifically captures firms’ differential exposure to marginal utility shocks. Our results complement (Whited and Wu, 2006) by providing direct evidence that financially constrained firms earn higher returns due to their amplified sensitivity to consumption disasters rather than purely idiosyncratic risks. Methodologically, we advance the literature by conducting comprehensive robustness tests following the protocol of (Novy-Marx and Velikov, 2023b), demonstrating that DER survives stringent hurdles including controls for 210 other anomalies and multiple transaction cost specifications. The DER strategy generates significant alpha of 22 basis points monthly even after controlling for six closely related anomalies and the Fama-French six factors simultaneously, establishing its incremental contribution to the investment opportunity set. These findings support consumption-based theories of the cross-section while providing practitioners with an implementable strategy that expands the mean-variance frontier net of trading costs, suggesting that markets efficiently price consumption risk exposure embedded in corporate financing decisions.

## 2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 6000-6999 and 4900-4999) following standard practice in the corporate finance literature. Debt Expansion Restraint signal relies on two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the dollar amount of long-term debt issued during the fiscal year, capturing new borrowing activities with maturities exceeding one year. This item reflects the

proceeds from issuing bonds, notes, and other long-term debt instruments during the reporting period. Total assets (AT) measures the book value of all assets reported on the balance sheet at fiscal year-end, providing a comprehensive measure of firm size and serving as a natural scaling variable for cross-sectional comparisons. We construct the Debt Expansion Restraint signal by calculating the year-over-year change in long-term debt issuance, scaled by lagged total assets. Specifically, we compute  $(DLTIS(t) - DLTIS(t-1)) / AT(t-1)$  for each firm-year observation. This measure captures the change in a firm's debt issuance activity relative to its asset base, with positive values indicating acceleration in debt financing and negative values suggesting deceleration or restraint in accessing debt markets. The scaling by lagged assets ensures comparability across firms of different sizes and mitigates potential mechanical correlations with contemporaneous firm characteristics. We calculate this signal using annual observations based on fiscal year-end values and require non-missing values for DLTIS in consecutive years and lagged assets to ensure data quality. Observations with zero or negative lagged assets are excluded from the analysis.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the DER signal. Panel A plots the time-series of the mean, median, and interquartile range for DER. On average, the cross-sectional mean (median) DER is -0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DER data. The signal's interquartile range spans -0.06 to 0.05. Panel B of Figure 1 plots the time-series of the coverage of the DER signal for the CRSP universe. On average, the DER signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

## 4 Does DER predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DER using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DER portfolio and sells the low DER portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DER strategy earns an average return of 0.26% per month with a t-statistic of 3.45. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.25% to 0.31% per month and have t-statistics exceeding 3.22 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 4.66 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 535 stocks and an average market capitalization of at least \$1,410 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 4.67. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-20bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.82. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DER trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DER strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DER, as well as average returns and alphas for long/short trading DER strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the DER strategy achieves an average return of 34 bps/month with a t-statistic of 3.65. Among these large cap stocks, the alphas for the DER strategy relative to the five most common factor models range from 25 to 36 bps/month with t-statistics between 2.64 and 3.90.

## 5 How does DER perform relative to the zoo?

Figure 2 puts the performance of DER in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the DER strategy falls in the distribution. The DER strategy’s gross (net) Sharpe ratio of 0.48 (0.37) is greater than 87% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DER strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the DER strategy would have yielded \$3.21 which ranks the DER strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DER strategy would have yielded \$1.92 which ranks the DER strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.



of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the DER relative to those. Panel A shows that the DER strategy gross alphas fall between the 58 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DER strategy has a positive net generalized alpha for five out of the five factor models. In these cases DER ranks between the 76 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does DER add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of DER with 210 filtered anomaly signals.<sup>3</sup> [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DER or at least to weaken the power DER has predicting the cross-section of returns. [Figure 7](#) plots histograms

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{DER}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DER trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DER and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DER signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on DER is -1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the DER strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DER strategy earns alphas that range from 22-24bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DER trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.01.

## 7 Does DER add relative to the whole zoo?

Finally, we can ask how much adding DER to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DER signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DER grows to \$1243.77.

## 8 Conclusion

This study provides compelling evidence that Debt Expansion Restraint (DER) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx

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<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DER is available.

and Velikov (2023), demonstrates that DER-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

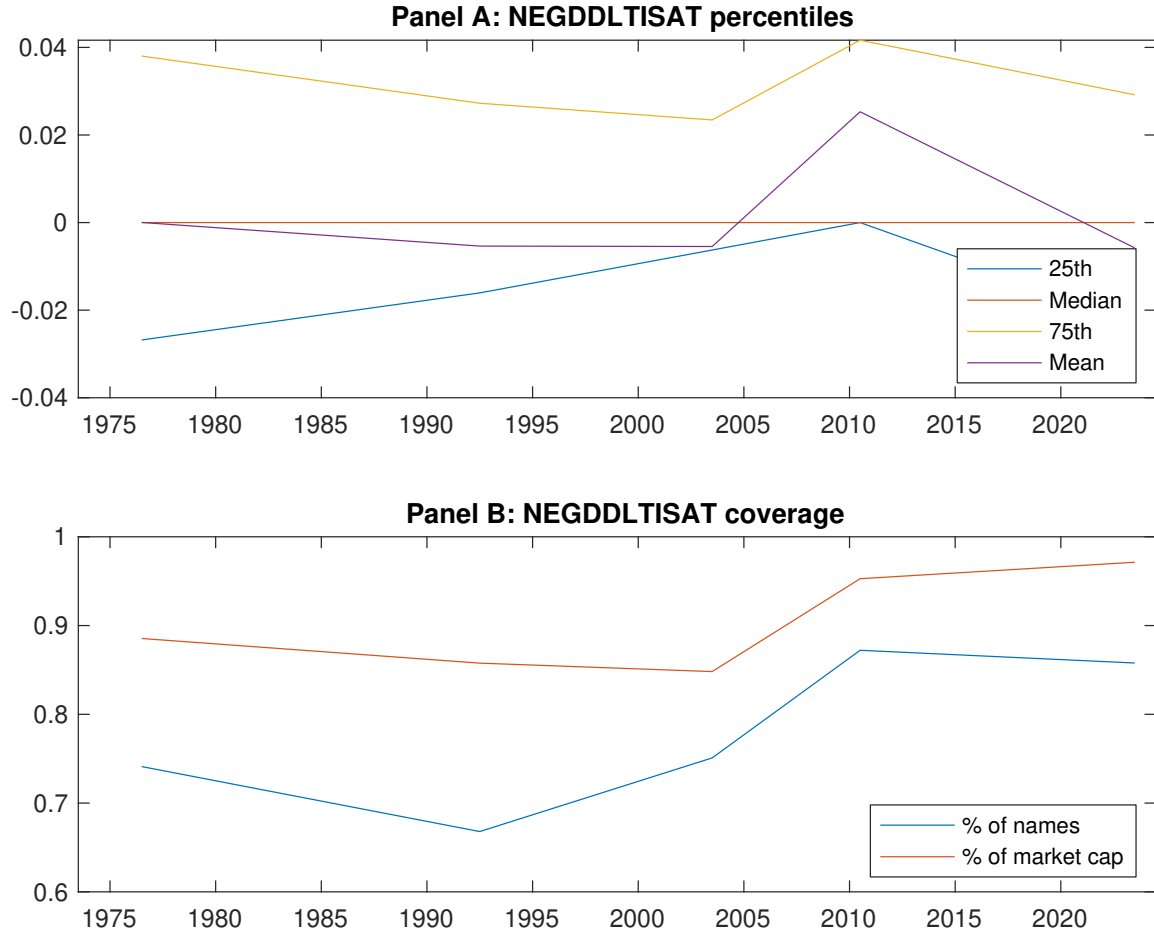
The key findings reveal that a value-weighted long/short portfolio sorted on DER delivers an annualized Sharpe ratio of 0.48 gross and 0.37 net of transaction costs, indicating that the signal remains profitable even after realistic implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 25 basis points per month ( $t$ -statistic = 3.22) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of a 22 basis point monthly alpha ( $t$ -statistic = 3.01) even after controlling for six closely related strategies including net debt financing, changes in financial liabilities, and various measures of asset growth and external financing.

These findings have important practical implications for both investors and corporate managers. For investors, DER provides a viable strategy for enhancing portfolio returns, particularly given its robustness to transaction costs and its incremental predictive power beyond existing factors. The signal’s economic magnitude and statistical significance suggest it captures a distinct dimension of firm behavior related to financial conservatism that is not fully reflected in current asset prices. For corporate managers, our results imply that markets reward firms exhibiting restraint in debt expansion, potentially reflecting investor preferences for financial prudence or concerns about agency costs associated with excessive leverage.

Several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe of stocks examined may influence the generalizability of our findings. Second, the underlying economic mechanism driving the DER anomaly remains partially unexplained, though it appears related to broader themes of financial constraint

and capital structure decisions. Third, the strategy’s performance during different market regimes and economic cycles deserves further investigation to fully understand its risk-return profile.

Future research should explore several promising avenues. First, investigating the international evidence on DER would help establish whether this anomaly is a global phenomenon or specific to certain markets. Second, examining the interaction between DER and firm characteristics such as size, profitability, and growth opportunities could provide deeper insights into when and why the signal is most effective. Third, developing a theoretical framework that explains why debt expansion restraint predicts returns would enhance our understanding of the underlying economic forces. Finally, investigating the time-varying nature of the DER premium and its relationship to macroeconomic conditions, credit market cycles, and aggregate leverage trends could yield valuable insights for both academic research and practical implementation. Despite these areas for future investigation, our results establish DER as a robust and economically meaningful predictor of stock returns that adds value beyond existing factor models and deserves consideration in both academic asset pricing research and practical portfolio construction.



**Figure 1:** Times series of DER percentiles and coverage. This figure plots descriptive statistics for DER. Panel A shows cross-sectional percentiles of DER over the sample. Panel B plots the monthly coverage of DER relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DER. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Excess returns and alphas on DER-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.59<br>[2.81]   | 0.64<br>[3.53]   | 0.62<br>[3.13]   | 0.72<br>[4.04]   | 0.85<br>[4.26]   | 0.26<br>[3.45]   |
| $\alpha_{CAPM}$                                                                   | -0.12<br>[-2.23] | 0.03<br>[0.56]   | -0.05<br>[-0.89] | 0.11<br>[2.49]   | 0.18<br>[3.11]   | 0.30<br>[4.02]   |
| $\alpha_{FF3}$                                                                    | -0.12<br>[-2.22] | -0.02<br>[-0.40] | -0.00<br>[-0.02] | 0.09<br>[2.02]   | 0.19<br>[3.49]   | 0.31<br>[4.10]   |
| $\alpha_{FF4}$                                                                    | -0.10<br>[-1.92] | 0.01<br>[0.12]   | 0.04<br>[0.82]   | 0.06<br>[1.31]   | 0.17<br>[3.05]   | 0.27<br>[3.58]   |
| $\alpha_{FF5}$                                                                    | -0.12<br>[-2.25] | -0.06<br>[-1.32] | 0.05<br>[0.92]   | 0.01<br>[0.15]   | 0.15<br>[2.65]   | 0.27<br>[3.53]   |
| $\alpha_{FF6}$                                                                    | -0.11<br>[-2.05] | -0.04<br>[-0.87] | 0.08<br>[1.51]   | -0.01<br>[-0.26] | 0.13<br>[2.41]   | 0.25<br>[3.22]   |
| Panel B: Fama and French (2018) 6-factor model loadings for DER-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 1.06<br>[83.62]  | 0.98<br>[96.26]  | 0.99<br>[77.85]  | 0.98<br>[97.21]  | 1.02<br>[79.29]  | -0.04<br>[-2.21] |
| $\beta_{SMB}$                                                                     | 0.16<br>[8.39]   | -0.14<br>[-9.08] | -0.02<br>[-0.87] | -0.04<br>[-2.72] | 0.14<br>[7.51]   | -0.01<br>[-0.55] |
| $\beta_{HML}$                                                                     | -0.03<br>[-1.21] | 0.16<br>[8.06]   | -0.10<br>[-4.23] | 0.02<br>[1.22]   | -0.14<br>[-5.53] | -0.11<br>[-3.16] |
| $\beta_{RMW}$                                                                     | 0.08<br>[3.16]   | 0.12<br>[5.75]   | -0.05<br>[-2.01] | 0.11<br>[5.53]   | 0.02<br>[0.82]   | -0.06<br>[-1.66] |
| $\beta_{CMA}$                                                                     | -0.09<br>[-2.35] | 0.00<br>[0.02]   | -0.11<br>[-2.90] | 0.16<br>[5.38]   | 0.15<br>[4.09]   | 0.24<br>[4.66]   |
| $\beta_{UMD}$                                                                     | -0.02<br>[-1.24] | -0.03<br>[-3.09] | -0.05<br>[-4.05] | 0.03<br>[2.78]   | 0.02<br>[1.46]   | 0.03<br>[1.95]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 671              | 535              | 1060             | 590              | 641              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1410             | 3195             | 2218             | 3114             | 1416             |                  |

**Table 2:** Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DER strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

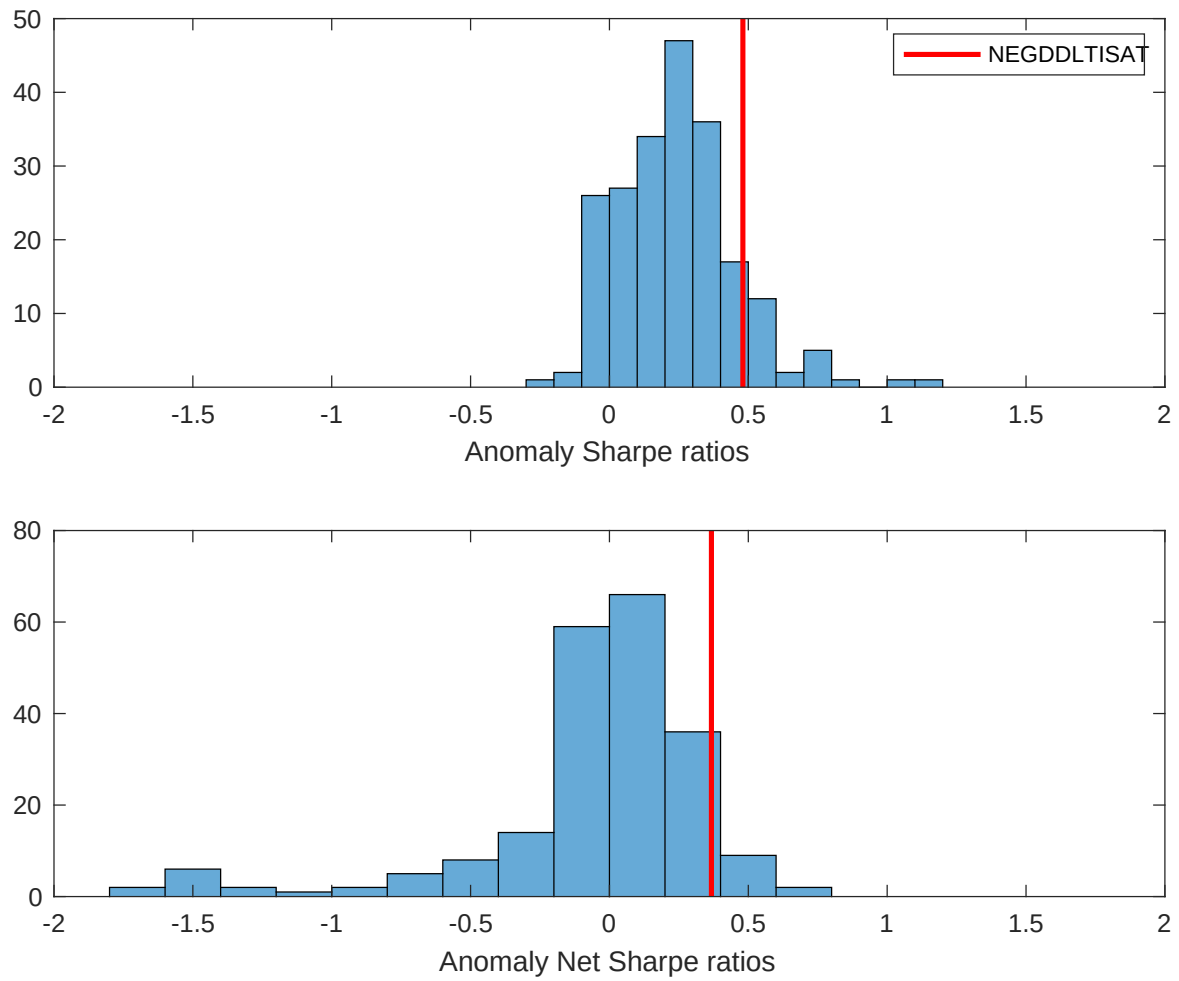
| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.26<br>[3.45]     | 0.30<br>[4.02]           | 0.31<br>[4.10]          | 0.27<br>[3.58]          | 0.27<br>[3.53]          | 0.25<br>[3.22]          |
| Quintile                                                                 | NYSE   | EW      | 0.21<br>[4.67]     | 0.24<br>[5.27]           | 0.22<br>[4.91]          | 0.20<br>[4.46]          | 0.21<br>[4.59]          | 0.20<br>[4.32]          |
| Quintile                                                                 | Name   | VW      | 0.25<br>[3.42]     | 0.29<br>[4.11]           | 0.30<br>[4.23]          | 0.26<br>[3.59]          | 0.25<br>[3.42]          | 0.22<br>[3.04]          |
| Quintile                                                                 | Cap    | VW      | 0.21<br>[3.29]     | 0.25<br>[3.82]           | 0.25<br>[3.78]          | 0.20<br>[3.10]          | 0.17<br>[2.70]          | 0.15<br>[2.29]          |
| Decile                                                                   | NYSE   | VW      | 0.23<br>[2.42]     | 0.31<br>[3.30]           | 0.28<br>[2.94]          | 0.24<br>[2.49]          | 0.12<br>[1.33]          | 0.11<br>[1.19]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.20<br>[2.64]     | 0.23<br>[3.13]           | 0.24<br>[3.17]          | 0.22<br>[2.91]          | 0.21<br>[2.76]          | 0.19<br>[2.54]          |
| Quintile                                                                 | NYSE   | EW      | -0.05<br>[-0.82]   |                          |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.19<br>[2.58]     | 0.24<br>[3.31]           | 0.24<br>[3.38]          | 0.22<br>[3.06]          | 0.20<br>[2.80]          | 0.18<br>[2.55]          |
| Quintile                                                                 | Cap    | VW      | 0.16<br>[2.48]     | 0.20<br>[3.06]           | 0.19<br>[2.98]          | 0.17<br>[2.63]          | 0.13<br>[2.09]          | 0.12<br>[1.81]          |
| Decile                                                                   | NYSE   | VW      | 0.16<br>[1.66]     | 0.24<br>[2.50]           | 0.21<br>[2.19]          | 0.19<br>[1.97]          | 0.08<br>[0.90]          | 0.07<br>[0.75]          |



**Table 3:** Conditional sort on size and DER

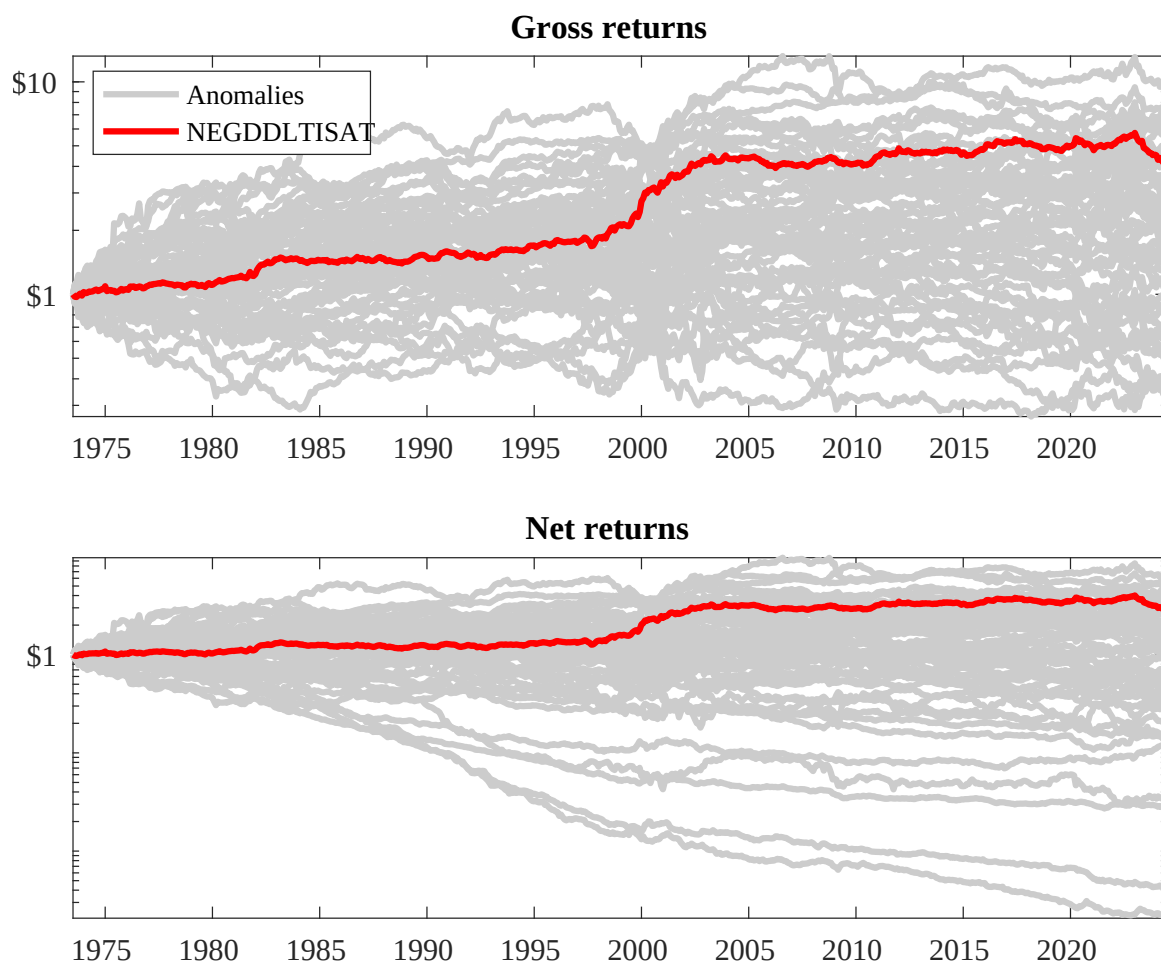
This table presents results for conditional double sorts on size and DER. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DER. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DER and short stocks with low DER. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | DER Quintiles |                |                |                |                | DER Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.65<br>[2.36] | 0.86<br>[3.20] | 0.94<br>[3.41] | 0.87<br>[3.18] | 0.83<br>[2.91] | 0.17<br>[1.86]                                     | 0.20<br>[2.16]  | 0.19<br>[1.99] | 0.14<br>[1.44] | 0.15<br>[1.62] | 0.12<br>[1.28] |
|                                                                       | (2)           | 0.72<br>[2.69] | 0.90<br>[3.65] | 0.82<br>[3.22] | 0.88<br>[3.58] | 0.87<br>[3.42] | 0.14<br>[1.78]                                     | 0.18<br>[2.26]  | 0.15<br>[1.96] | 0.16<br>[1.98] | 0.14<br>[1.73] | 0.14<br>[1.78] |
|                                                                       | (3)           | 0.78<br>[3.06] | 0.79<br>[3.62] | 0.81<br>[3.35] | 0.85<br>[3.79] | 0.90<br>[3.83] | 0.12<br>[1.52]                                     | 0.19<br>[2.41]  | 0.18<br>[2.28] | 0.14<br>[1.75] | 0.16<br>[1.98] | 0.13<br>[1.62] |
|                                                                       | (4)           | 0.73<br>[3.18] | 0.78<br>[3.75] | 0.84<br>[3.84] | 0.74<br>[3.55] | 0.91<br>[4.15] | 0.17<br>[2.31]                                     | 0.21<br>[2.75]  | 0.20<br>[2.68] | 0.18<br>[2.28] | 0.19<br>[2.39] | 0.17<br>[2.12] |
|                                                                       | (5)           | 0.50<br>[2.56] | 0.60<br>[3.36] | 0.58<br>[2.94] | 0.61<br>[3.35] | 0.83<br>[4.42] | 0.34<br>[3.65]                                     | 0.36<br>[3.88]  | 0.36<br>[3.90] | 0.29<br>[3.15] | 0.29<br>[3.13] | 0.25<br>[2.64] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | DER Quintiles |                |                |                |                | DER Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 392            | 393            | 393            | 393            | 389            | 37                                                 | 34              | 32             | 33             | 35             |                |
|                                                                       | (2)           | 107            | 107            | 107            | 108            | 107            | 61                                                 | 61              | 60             | 61             | 61             |                |
|                                                                       | (3)           | 76             | 77             | 76             | 76             | 77             | 107                                                | 110             | 105            | 107            | 108            |                |
|                                                                       | (4)           | 64             | 65             | 64             | 65             | 64             | 229                                                | 240             | 229            | 235            | 230            |                |
| (5)                                                                   | 59            | 59             | 59             | 59             | 59             | 1453           | 2220                                               | 1800            | 2187           | 1517           |                |                |



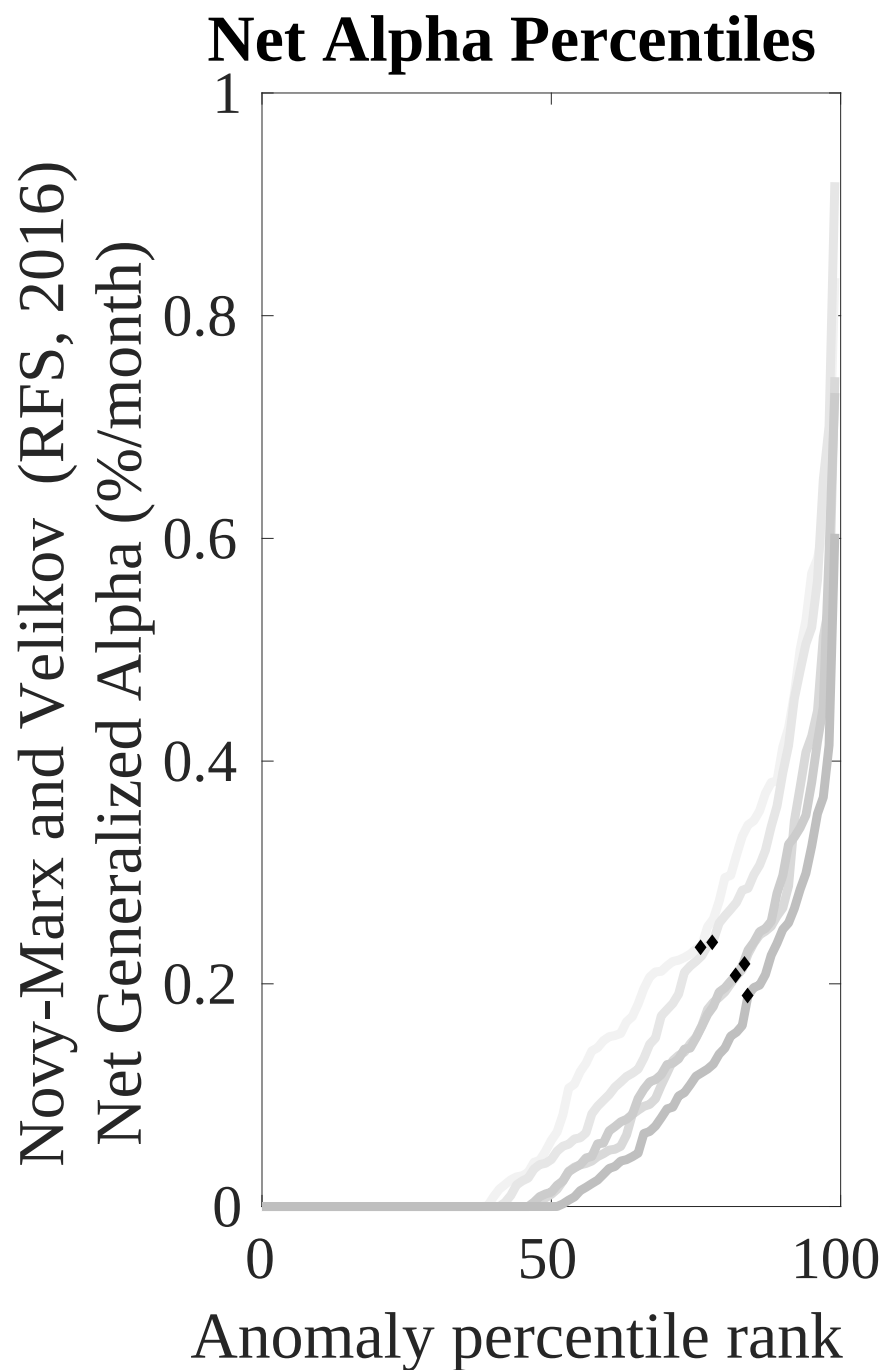
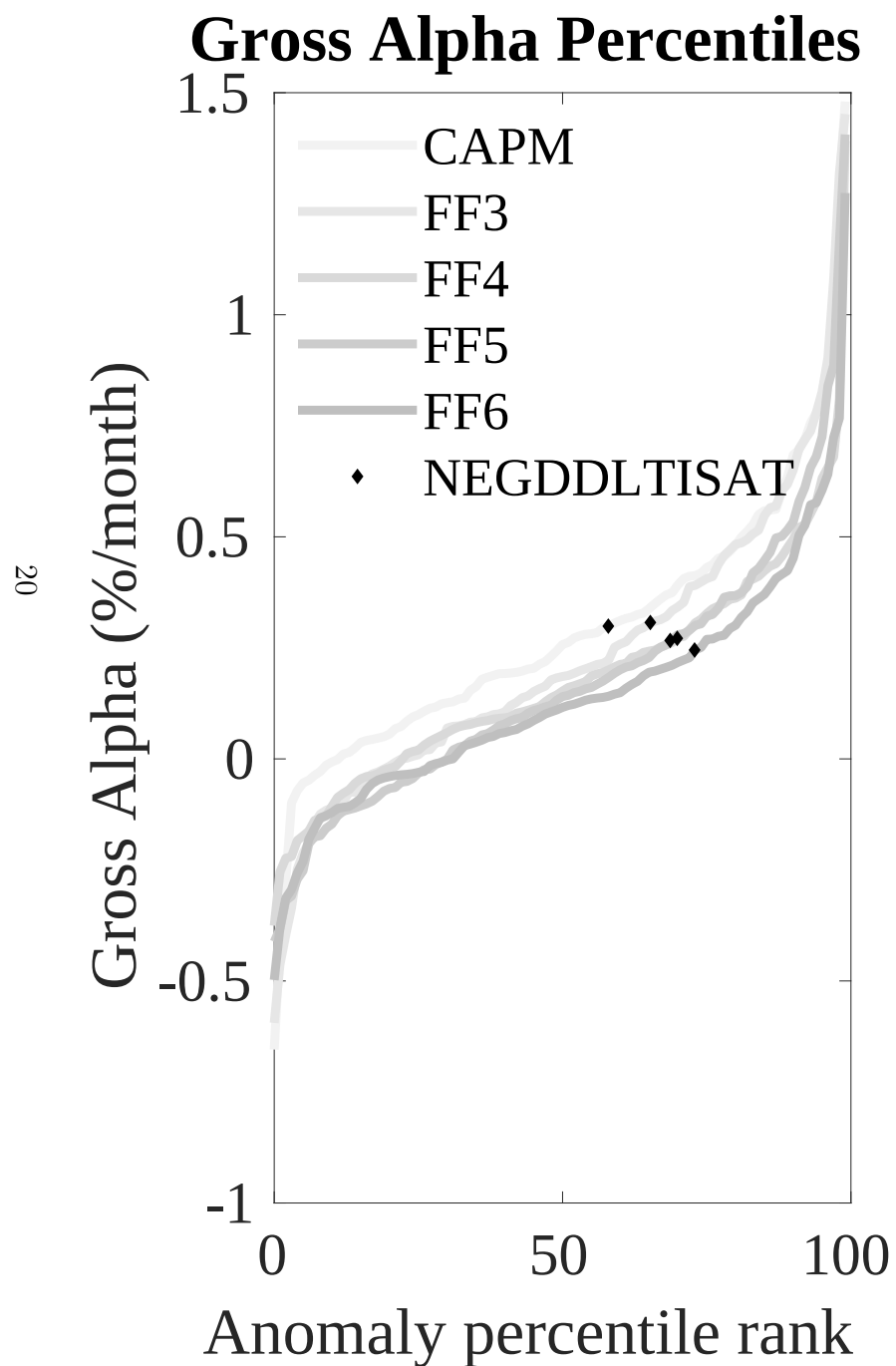
**Figure 2:** Distribution of Sharpe ratios.

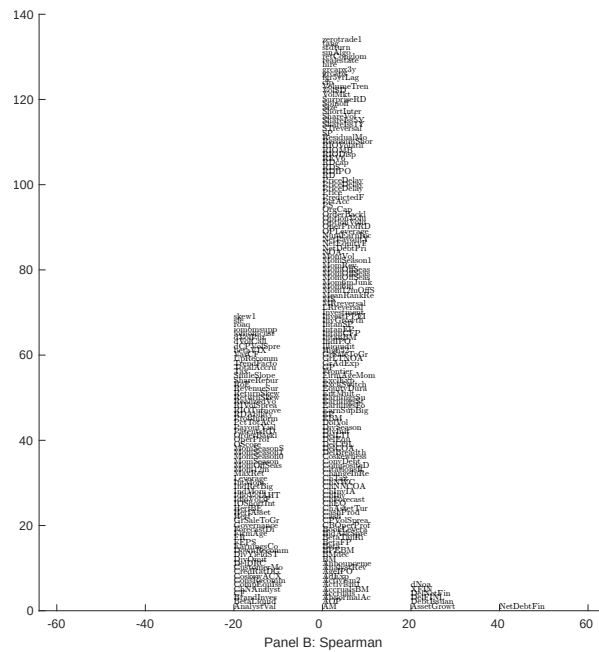
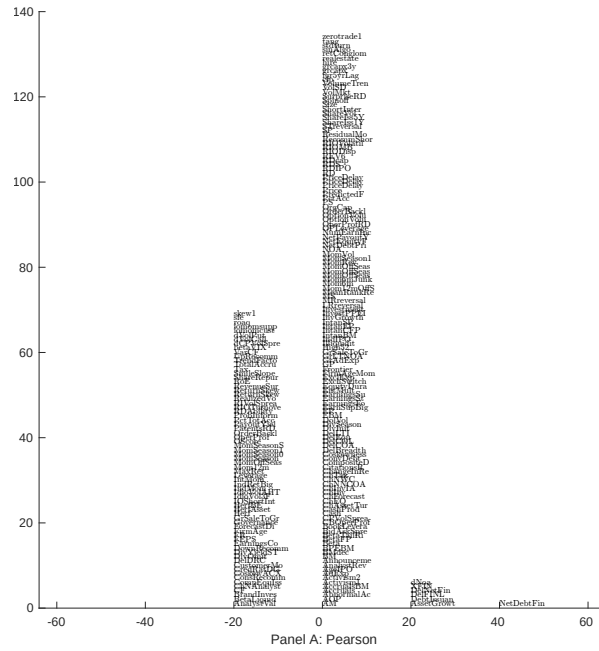
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DER with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

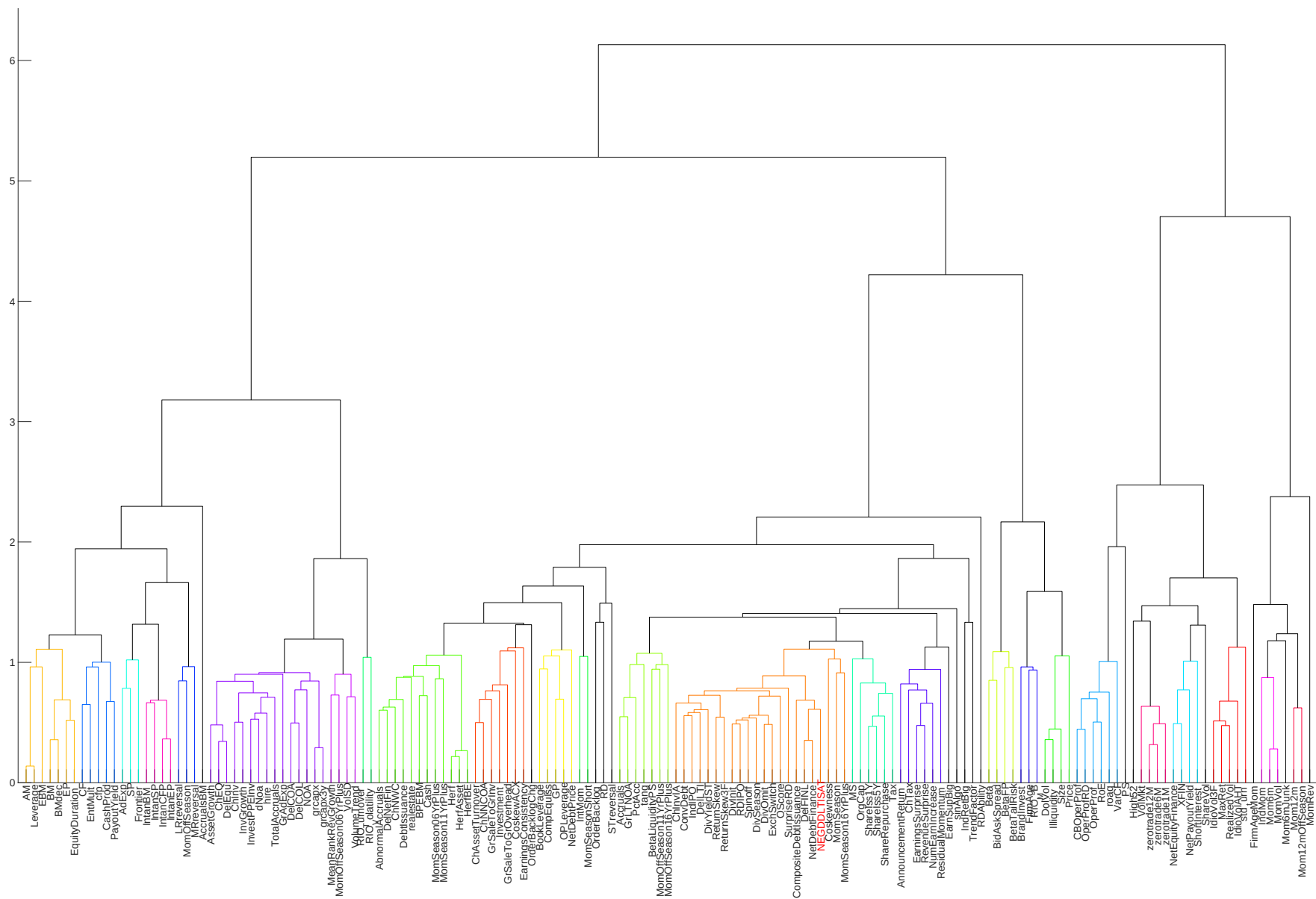
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DER trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.





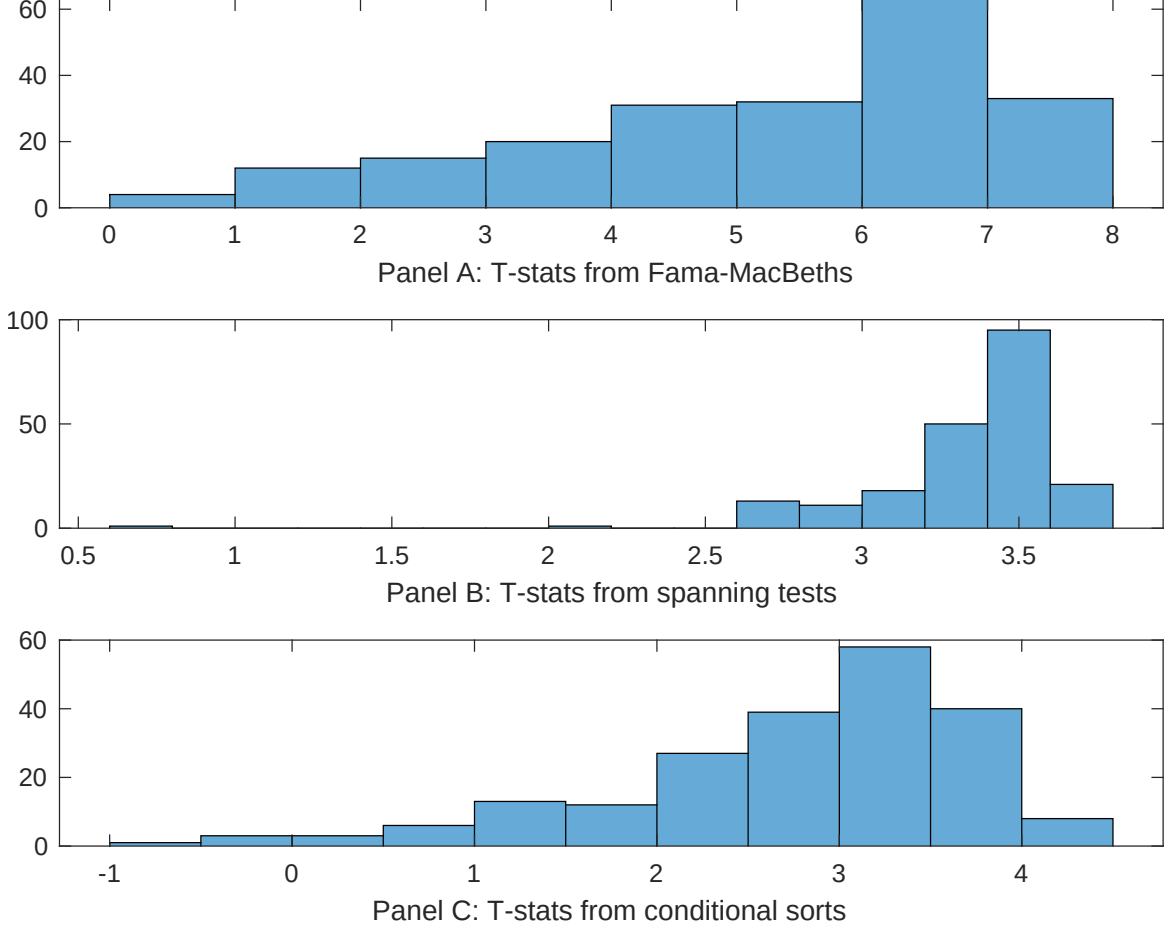
**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DER. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{DER}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DER trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DER. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

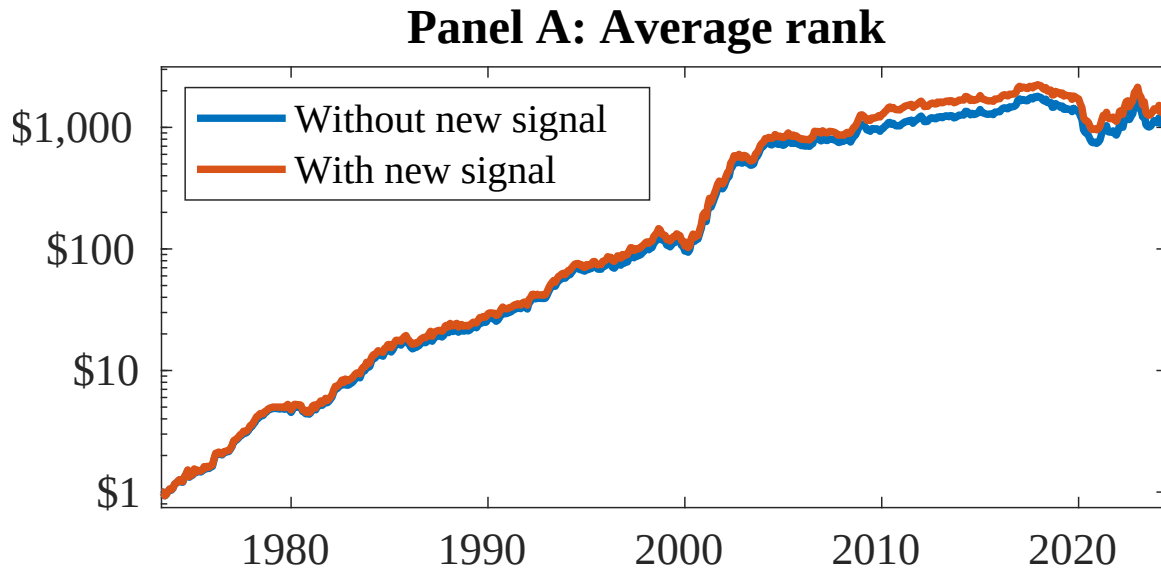
|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.13<br>[5.33] | 0.13<br>[5.36] | 0.14<br>[5.32] | 0.14<br>[5.69] | 0.14<br>[5.73] | 0.14<br>[5.65] | 0.15<br>[5.75]   |
| DER             | 0.12<br>[1.21] | 0.10<br>[1.11] | 0.51<br>[5.11] | 0.20<br>[1.88] | 0.21<br>[2.33] | 0.29<br>[0.31] | -0.14<br>[-1.22] |
| Anomaly 1       | 0.20<br>[8.54] |                |                |                |                |                | 0.11<br>[1.59]   |
| Anomaly 2       |                | 0.18<br>[9.14] |                |                |                |                | -0.10<br>[-2.20] |
| Anomaly 3       |                |                | 0.40<br>[6.87] |                |                |                | 0.21<br>[0.35]   |
| Anomaly 4       |                |                |                | 0.19<br>[6.09] |                |                | 0.11<br>[2.16]   |
| Anomaly 5       |                |                |                |                | 0.17<br>[7.67] |                | 0.69<br>[2.29]   |
| Anomaly 6       |                |                |                |                |                | 0.14<br>[9.71] | 0.96<br>[4.63]   |
| # months        | 612            | 612            | 612            | 612            | 612            | 612            | 612              |
| $\bar{R}^2(\%)$ | 0              | 0              | 0              | 1              | 0              | 0              | 0                |



**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DER trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{DER} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

|                 |                  |                  |                   |                   |                   |                   |                   |
|-----------------|------------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Intercept       | 0.22<br>[2.94]   | 0.22<br>[2.88]   | 0.24<br>[3.20]    | 0.22<br>[2.89]    | 0.23<br>[3.12]    | 0.23<br>[2.99]    | 0.22<br>[3.01]    |
| Anomaly 1       | 19.46<br>[4.77]  |                  |                   |                   |                   |                   | 10.38<br>[1.87]   |
| Anomaly 2       |                  | 16.73<br>[3.94]  |                   |                   |                   |                   | 10.42<br>[1.79]   |
| Anomaly 3       |                  |                  | 10.13<br>[3.46]   |                   |                   |                   | 9.19<br>[2.99]    |
| Anomaly 4       |                  |                  |                   | 13.69<br>[3.64]   |                   |                   | 9.30<br>[2.36]    |
| Anomaly 5       |                  |                  |                   |                   | 9.28<br>[2.72]    |                   | 4.22<br>[1.12]    |
| Anomaly 6       |                  |                  |                   |                   |                   | 4.34<br>[1.00]    | -8.15<br>[-1.68]  |
| mkt             | -3.52<br>[-2.05] | -3.17<br>[-1.84] | -3.74<br>[-2.16]  | -1.68<br>[-0.93]  | -3.68<br>[-2.12]  | -3.54<br>[-2.03]  | -2.26<br>[-1.27]  |
| smb             | -2.96<br>[-1.13] | -2.96<br>[-1.12] | -0.13<br>[-0.05]  | 3.16<br>[1.10]    | -1.21<br>[-0.46]  | -1.16<br>[-0.44]  | 0.63<br>[0.22]    |
| hml             | -9.73<br>[-2.94] | -9.41<br>[-2.82] | -10.85<br>[-3.25] | -9.00<br>[-2.68]  | -11.52<br>[-3.42] | -10.88<br>[-3.21] | -8.46<br>[-2.52]  |
| rmw             | -7.67<br>[-2.26] | -7.04<br>[-2.07] | -4.06<br>[-1.19]  | -13.71<br>[-3.37] | -5.36<br>[-1.57]  | -5.46<br>[-1.60]  | -11.89<br>[-2.93] |
| cma             | 19.28<br>[3.76]  | 18.73<br>[3.57]  | 15.33<br>[2.69]   | 15.40<br>[2.74]   | 16.98<br>[2.95]   | 21.12<br>[3.51]   | 6.63<br>[1.02]    |
| umd             | 2.34<br>[1.36]   | 2.14<br>[1.22]   | 2.55<br>[1.46]    | 3.50<br>[2.04]    | 3.57<br>[2.07]    | 3.45<br>[1.98]    | 1.26<br>[0.72]    |
| # months        | 612              | 612              | 612               | 612               | 612               | 612               | 612               |
| $\bar{R}^2(\%)$ | 10               | 9                | 9                 | 9                 | 8                 | 7                 | 13                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DER. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

## References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gomes, J. F. and Schmid, L. (2016). Financing investment. *Journal of Finance*, 71(6):2717–2752.
- Livdan, D., Sapriz, H., and Zhang, L. (2009). Financially constrained stock returns. *Journal of Finance*, 64(4):1827–1862.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Review of Financial Studies*, 36(3):1103–1145.
- Whited, T. M. and Wu, G. (2006). Financial constraints risk. *Review of Financial Studies*, 19(2):531–559.